Episode 9: The Pitch Deck

Programming with Python · Tutorial 09

The pitch meeting is Friday, and the investor gave one instruction: **“Bring me charts I can’t argue with.”** Last week you turned the data room into numbers she could trust; this week those numbers have to be *seen*. Same file, `orders_messy.csv`, now with pictures: you pick the chart that fits each question, save it as a picture file, and end the lab with a deck of four that tells one story. Tobi has already “built the deck”: two slides and one argument for a new rider. Your job is to **supervise**: catch the chart she can argue with before she does.

Run the lab on your laptop

The lab starts once your laptop has passed the git check from the lecture: `git --version`, `gh auth status` and `git config --global user.email` in Zed’s terminal, shown to your lecturer. That check is the one thing you have to show before you leave today. Missed it? The steps are in the one-time setup on the Git Basics page.

The lab is a Python script again. It runs in the `python-labs` folder, next to last week’s files.

1. Download the script and move it into your `python-labs` folder. `orders_messy.csv` is still there from last week; if not, download it again:

   Lab script Orders data

2. In Zed, open the `python-labs` folder, then the terminal (`Ctrl+`).
3. Run `uv run python lab_09_pitch.py`.

> **No Zed on your laptop, or only an iPad?**
> Use a **GitHub codespace**: a Linux computer that runs in a browser tab.
>
> 1. Sign in at github.com/codespaces and open last week’s codespace again. If you have none, start the **Blank** template and follow the uv guide (the macOS or Linux lines) in its terminal, down to `uv add pandas matplotlib`.
> 2. Go into the folder with `cd python-labs` and download the script. The second line is only needed in a new codespace:
>
>    ```
>    curl -O https://python.tobiasvlcek.com/labs/lab_09_pitch.py
>    curl -O https://python.tobiasvlcek.com/labs/orders_messy.csv
>    ```
>
> 3. Run `uv run python lab_09_pitch.py`.
>
> To look at a chart, click its PNG file in the file list on the left. A codespace has no Zed, so keep the chatbot open in a second tab for AI help.

The first run stops with `AssertionError: Task 1: not attempted yet.`, as it did last week: the script checks every task and stops at the first one you have not solved yet. The very first run can take a little longer, because matplotlib sets up its fonts once.

If the run ends in `ModuleNotFoundError` instead, the terminal is not inside `python-labs`, or the packages are missing. In that folder, run `uv add pandas matplotlib` once.

A script has no cell that could show a chart, so every chart goes into a picture file in `python-labs`. Task 2 in the script shows the three lines that end every chart. To look at a picture, click its file in Zed's project panel, the file list on the left.

> ! **Look at your charts**
> The checks only test the form of a chart: that it has a title, that its bars add up to your own numbers. They cannot see whether the chart tells the truth. And **two things in Tobi's deck are wrong without any task saying so**: one is in his numbers, one is in what he claims. No check catches either of them.

The investor gets four slides, not six: Task 8 asks which four of your charts tell her one story, and in which order. When every check passes, the script puts those four into `pitch_deck.png`, in your order, and prints a block that starts with `SHOW THIS IN CLASS`. Show both to your lecturer. As in lab 08, that is feedback, not a condition for leaving, and a lab you finish at home does not have to be shown later.

Checkpoint 5 next week is still handed in from the browser, and the script labs no longer practise that. So take two minutes before you go: open today's last exercise, ex_09_d, press **Cmd/Ctrl+S**, then menu → Download → **Download Python code**, and find the file in your Downloads folder.

Optional practice for Task 8: ex_09_e puts four slides in story order and finds the one that breaks the color rule.

Bibliography